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Potential overestimation of carbon dioxide emissions from croplands on organic soils in cool temperate and boreal regions based on a case study from Norway

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Peatlands drained for agriculture and other uses release substantial carbon dioxide. Many countries estimate these emissions using the 2014 IPCC Tier 1 emission factors. Here we calibrated an ecosystem model with data from two cultivated peatland sites in Norway and simulate carbon dioxide emissions at 50 sites nationwide for 2001–2022. Model results showed that carbon dioxide emissions were strongly controlled by water table depth and aligned well with observations from other European peatlands of similar climate zones. Crucially, the Tier 1 emission factor matched our simulations only under very deep water tables (< –0.7 m), but overestimated emissions by 31–88% when water levels ranged from –0.7 m to –0.3 m. This indicates that Tier 1 methods may overstate emissions from cultivated peatlands in cool temperate and boreal regions, inflating estimates of mitigation potential. Tier 2 or 3 approaches can reduce uncertainty but require more field data.

Peatlands account for a substantial fraction of global soil carbon (C) stock, due to their anaerobic soil environment fostering a slow C decomposition process. Since the 17th Century, extensive areas of global peatlands have been drained for agriculture, forestry, peat extraction, and urban development[1]. This practice, notably widespread in Europe, exposes peat C to an aerobic soil environment that accelerates C decomposition. Consequently, drained peatland can rapidly release substantial amounts of C into the atmosphere as carbon dioxide (CO_2), a potent greenhouse gas (GHG)[2–4].

In Norway, peat depth varies across regions, with an average value of approximately 1.7 m and can occasionally reach 10–12 m[5]. Drainage of peatlands for cultivation began in the 1750 s on a large scale with government support[6]. Carbon content in these peat soils was estimated to be on average ~45% with a stock of 135 t C ha^{-1} [7]. Agricultural drainage typically features perforated pipes buried at 0.8 m depth with lateral spacing of 7–8 m[8]. Only after the 1970s, when the government began to phase out subsidies, were drainage activities largely reduced. Subsequently, the government implemented several regulations that restrict peatland drainage in Norway, but most existing drainages are still actively maintained.

Currently, drained peatland used to grow crops covers ~67 kha (0.21% of Norwegian land area). These areas were estimated to have released a total amount of 1753 kt CO_2 in 2023, representing the highest emission among all categories within the Land Use, Land-Use Change and Forestry (LULUCF) sector[9]. These reported emissions are estimated following the "Tier 1 methodology", which is based on the default emission factors (EFs) published by IPCC[10]. The Norwegian national report uses the EFs from "boreal and temperate cropland on organic soils", following IPCC climate zone classifications[11], to estimate GHG emissions from cultivated peatland in Norway. The Tier 1 method is widely used for GHG accounting in many countries since the method is mandatory to implement when countries lack field measurements to develop national specific emission factors ("Tier 2"). However, these Tier 1 EFs are calculated based on observations from a few studies worldwide in the corresponding climate zone and land-use type without accounting for temporal and spatial

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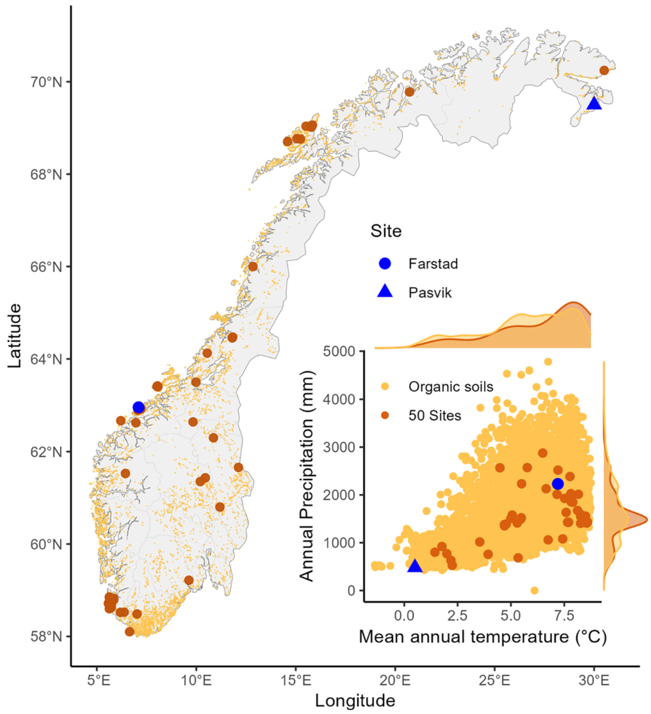

Fig. 1 | Map of spatial distribution of agricultural organic soils in Norway. Orange points represent the distribution of organic soils. Brown points are the 50 random sites, and the blue point is the experiment sites in Farstad and Pasvik. Inset displays annual precipitation against mean annual temperature (2001–2022) for the locations.

variances. Thus, the estimates are usually associated with large uncertainties, which are not incorporated into national GHG accounting. Due to the contrasting climatic and soil conditions in Norway, there is a need to investigate the spatial variability of GHG emissions across different regions of the country and compare them with the Tier 1 EFs to gauge the magnitude of uncertainties. Nonetheless, the lack of field-based GHG measurements[12] remains a major obstacle for improving national accounting methodologies for peatlands in Norway.

Hydrology, particularly water table depth, is the most important factor that determines the GHG emissions from a peatland[13]. Raising water tables in drained peatlands can restore anaerobic conditions that reduce the decomposition rate and the CO_2 emissions[14,15]. At the same time, raising water tables can also favor methane (CH_4) production, leading to increased CH_4 emissions, which can partially or even fully offset the reduction in CO_2 emissions[14]. Therefore, the success of raising water tables depends on the combined emissions of multiple GHGs, including CO_2, CH_4 and nitrous oxide (N_2O). Additionally, harvested biomass is removed from the ecosystem as an important C export, which should also be considered in C balance estimates for sustainable management evaluations.

To assess the C balance, this study focuses on modeling CO_2 emissions and net C budget from cultivated peatlands in Norway. Several models have been used for CO_2 flux simulations in peatlands, such as satellite-based GPP models[16], ECOSSE[17], Peatland-VU-NUCOM (PVN)[18], and the CoupModel[19]. While satellite-based models are limited by their inability to capture belowground processes, process-based models like ECOSSE and PVN have improved simulation of water table and vegetation dynamics. However, they still lack fully coupled ecohydrology and carbon dynamics. CoupModel is a comprehensive model that is designed to simulate water, heat, and mass transport within the soil-plant-atmosphere system. It has shown good performance in modeling GHG fluxes from peatland ecosystems[19–21]. In this study, we calibrated the CoupModel against data collected at two cultivated peatlands in northern and western Norway. Then, we used the calibrated model to estimate the CO_2 emissions at 50 randomly chosen locations on the map of croplands on organic soils in Norway. Here, we seek to answer the following questions:

(1) How effectively does the CoupModel capture CO_2 dynamics across sites with varying water table levels and climate conditions in Norway?

(2) Is the Tier 1 EF comparable to the simulated emissions over a wide range of climate conditions across Norway?

(3) How does raising the water table influence CO_2 fluxes from cultivated peatlands under different climates based on modeling simulations?

Results

Representativeness of the 50 random sites

Agricultural organic soils in Norway are predominantly distributed across the southern and central regions, extending northward along the coast. While scattered throughout, there is a notable concentration in the south, particularly around the 58°N to 62°N latitude range. The selected 50 random sites represent a mean annual temperature range from 1.2 to 9 °C and an annual precipitation range from ~400 to 2500 mm, covering >95% of locations with cultivated organic soils in Norway (Fig. 1). As the selection was weighed by the total area of cultivated peat in each zone, a greater number of sites were sampled in the south and west of the country where most of the cultivated organic soils are located. The experiment site in Farstad is in the warm and wet area while Pasvik is at the cold and dry end of all the locations, representing a wide range of climate conditions in Norway.

Modeled CO_2 fluxes at the Farstad and Pasvik sites

At the Farstad site, CoupModel performance varied across flux components during 2022–2024. For NEE (Fig. 2A), the model generally captured the direction of C exchange but underestimated net C uptake by 2 g C m^{-2} day^{-1} compared to measurements (Table 1). The scatter plot shows considerable spread around the 1:1 line, reflected in a moderate R^2 of 0.40 and an RMSE of 10.63. The modeled GPP showed similar underestimation as NEE compared to the measurements, with similar variability ($R^2 = 0.45$; RMSE = 11.82, Fig. 2B). In contrast, R_{eco} (Fig. 2C) was simulated with higher accuracy, closely matching measured values and showing the strongest correlation ($R^2 = 0.62$) and lowest RMSE (5.42). Overall, the model reproduced patterns of CO_2 fluxes at Farstad that are in line with the measured fluxes.

At the Pasvik site, the modeled CO_2 fluxes showed a better agreement with measured values in the wet plots (Fig. 3C) with lower errors (both RMSE and MAE) and higher R^2 compared to the dry and medium plots (Fig. 3A, B, Table 1). While the mean CO_2 fluxes were the highest (as C sources) in the dry plots (modeled: 3.9 or measured: 3.7 g C m^{-2} day^{-1}), followed by the medium plots (modeled: 0.34 or measured: 0.2 g C m^{-2} day^{-1}), the wet plots had CO_2 fluxes of −0.63 (modeled) or −0.95 (measured) g C m^{-2} day^{-1} as C sinks. For the cumulative fluxes of CO_2, the dry and medium plots were sources with emissions of ~1600 and 150 g C m^{-2}, respectively (Fig. 3A, B) whereas the wet plots were a small sink (~−450 g C m^{-2}) (Fig. 3C). The modeled cumulative CO_2 fluxes generally reflected similar sink/source patterns but showed the best agreement with the measured values in the dry plots but not in the medium and wet plots. In the medium plots, the large CO_2 uptake at the beginning of the growing season was underestimated by the model, while the CO_2 emission during the peak growing season was overestimated, making the total budget at the end of the season somehow still comparable. In the wet plots, the discrepancies between modeled and measured fluxes were present at the beginning of the season when measured CO_2 uptake was higher than the modeled values, resulting in underestimated C sink according to the model outputs.

The model partitioned NEE into GPP and R_{eco}. At the Farstad site, a similar seasonal pattern is present for 2023 and 2024 (Fig. 4), with fluxes remaining near zero during winter and increasing sharply from early spring (around day 100) through mid-summer. GPP consistently exceeds respiration during the peak growing season, creating periods of net CO_2

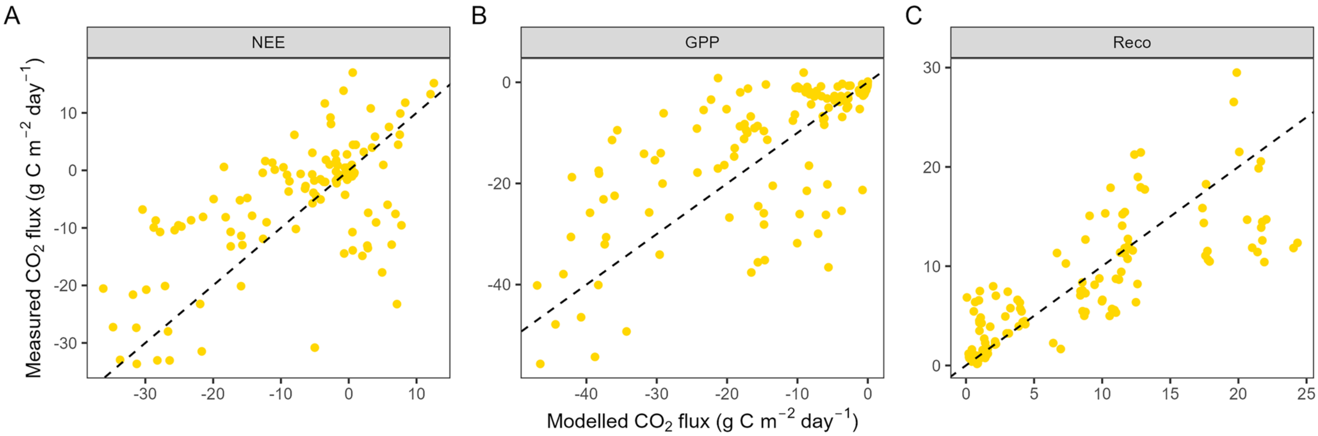

Fig. 2 | Performance of CoupModel at predicting CO_2 fluxes at Farstad site in 2022–2024. Scatter plots show modeled versus measured flux estimates (dashed lines indicate the 1:1 ratio) for NEE (**A**), GPP (**B**) and R_{eco} (**C**).

Table 1 | Statistics of comparing CoupModel modeled and measured NEE, GPP and R_{eco} at Farstad in 2022–2024, and NEE across plots with different water table levels at Pasvik in 2022

Site	Plot	Flux type	Modeled mean (g C m^{-2} day^{-1})	Measured mean (g C m^{-2} day^{-1})	N	RMSE	MAE	R^2
Farstad		NEE	−5.31	−7.37	121	10.63	−2.06	0.40
		GPP	−13.00	−14.96	121	11.82	1.97	0.45
		R_{eco}	7.99	8.19	123	5.42	0.20	0.62
Pasvik	Dry	NEE	3.90	3.70	407	3.95	3.06	0.57
	Medium	NEE	0.34	0.20	564	3.96	2.81	0.56
	Wet	NEE	−0.63	−0.95	463	2.77	2.03	0.67

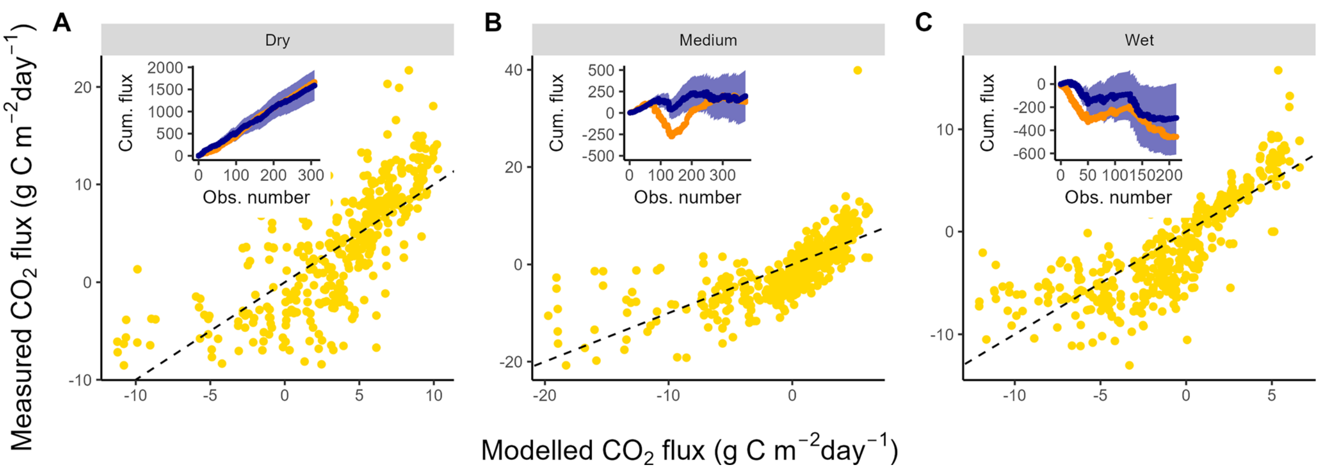

Fig. 3 | Performance of CoupModel at predicting CO_2 fluxes across plots with different water table levels at Pasvik in 2022. Scatter plots show modeled versus measured flux estimates (dashed lines indicate the 1:1 ratio) in the dry (**A**), medium (**B**) and wet plots (**C**). Inserts depict cumulative flux estimates (g C m^{-2}, blue points are modeled values with 95% confidence interval bands, and orange are measured values) as a function of sequential measurements (at 8 h intervals) to demonstrate temporal adherence. Note that the period (July 20 - August 10) with missing measurements was not considered in the cumulative values.

uptake. By the end of the year, the ecosystem reached CO_2 neutral in 2023 and slightly higher productivity in 2024.

For the Pasvik site, in the dry plots, the overall cumulative NEE is mainly dominated by respiration (Fig. 5A). By the end of the season, CO_2 uptake via photosynthesis accounted for <10% of the CO_2 released through respiration. By contrast, GPP was much higher in the medium plots and ~3/4 of the CO_2 release via respiration was compensated for by CO_2 uptake through photosynthesis (Fig. 5B). In the wet plots, the amount of GPP exceeded the respiratory emission after the peak growing season, turning the plots into C sinks (Fig. 5C).

Simulated CO_2 fluxes at the 50 sites across Norway

The simulated net CO_2 emissions showed a sharp reduction as water level changed from −1 m to −0.1 m (Fig. 6). Temperature played a crucial role in determining the magnitude of the fluxes and the emissions were lower in warmer regions. While the regions with mean annual temperature of 0–2 °C remained as CO_2 sources when water level reached −0.1 m, warmer regions with temperature of >6 °C became CO_2 sinks when water level was higher than −0.6 m. Simulated fluxes exhibited slight differences in their distributions when using parameter sets calibrated for the two sites; however,

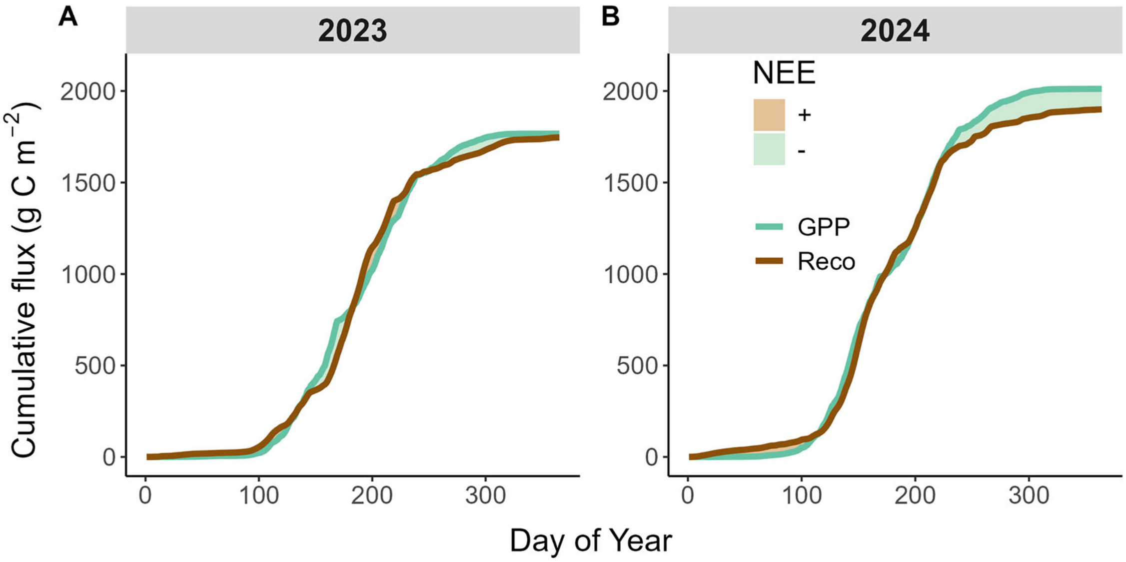

Fig. 4 | Modeled gross primary production (GPP) and ecosystem respiration (R_{eco}), showing cumulative estimates at Farstad. Net ecosystem exchanges (NEE) are indicated by the areas between GPP and R_{eco} (brown: net CO_2 source, green: net CO_2 sink) in 2023 (**A**) and 2024 (**B**).

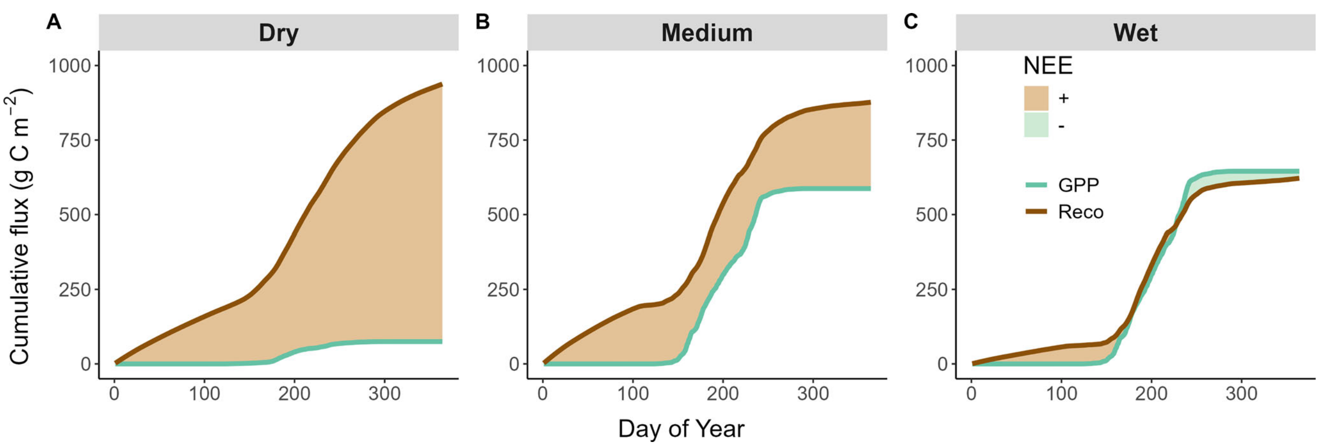

Fig. 5 | Modeled gross primary production (GPP) and ecosystem respiration (R_{eco}), showing cumulative estimates at Pasvik site in 2022. Net ecosystem exchanges (NEE) are indicated by the areas between GPP and R_{eco} (brown: net CO_2 source, green: net CO_2 sink) across plots with different water table levels (**A**: dry, **B**: medium, **C**: wet).

overall patterns remained consistent across water table levels and temperature gradients.

Among the tested key parameters in the model, humus decomposition potential (k_h) exhibited the most obvious effect on CO_2 fluxes (Fig. 7). By changing the k_h value from 0.0005 to 0.001 day^{-1}, CO_2 emissions increased by up to 3 folds at water level of −0.85 m. In addition, CO_2 fluxes were also sensitive to light use efficiency, where increasing the value from 2 to 3 g/MJ could significantly increase the CO_2 uptake via photosynthesis and made most sites CO_2 sinks regardless of water levels. Other parameters (harvest frequency and C/N ratio) showed only slight impacts on the overall CO_2 emissions. Compared to CO_2 fluxes that were observed from other studies, our simulated CO_2 fluxes were generally in line with them (Fig. 7). Although our simulations may seem to overestimate emissions at the low water table level of −0.85 m, the limited observations within this range also introduce substantial uncertainty in the measured fluxes. Simulated emissions were slightly greater when using parameter sets calibrated for Farstad than those for Pasvik; however, overall patterns remained consistent across water table levels (Fig. S3).

Net ecosystem carbon balance comparisons

Considering parameter sensitivity, model simulations indicate a clear increase in NECB with deeper water tables: mean values rise from 4.2 t C ha^{-1} yr^{-1} at water levels between −0.3 and −0.5 m to 7.7 t C ha^{-1} yr^{-1} below water table of −0.7 m (Fig. 8; Table 2). The Tier 1 EF of 7.9 t C ha^{-1} yr^{-1}, recommended for water tables lower than −0.3 m, closely matches our estimates for depths below −0.7 m and is 31–88% higher than simulated values for water levels between −0.7 and −0.3 m. Observations from Pasvik plots follow the same trend: the dry plot (mean water table −1.1 m) shows an NECB of 8.8 t C ha^{-1} yr^{-1} (95% CI: 6.1–11.4), comparable to the Tier 1 EF, whereas the medium plot (mean water table −0.4 m) has a lower NECB of 4.9 t C ha^{-1} yr^{-1} (95% CI: 3.2–6.5).

Discussion

This study used the process-based CoupModel to simulate CO_2 fluxes and NECB in drained cultivated peatlands from different regions of Norway. CO_2 emissions were strongly influenced by water table depth, with model outputs being generally consistent with observations from other studies. When comparing the Tier 1 CO_2 EF to the simulated NECB, we found the EF only matched our simulations under very deep-water tables (< −0.7 m), while overestimated emissions by 31–88% for sites with water levels between −0.7 m and −0.3 m. This result is also supported by field observations at Pasvik (Table 2). According to IPCC guidelines[10], the Tier 1 EF is applied to sites with water table depths below −0.3 m. However, given the substantial

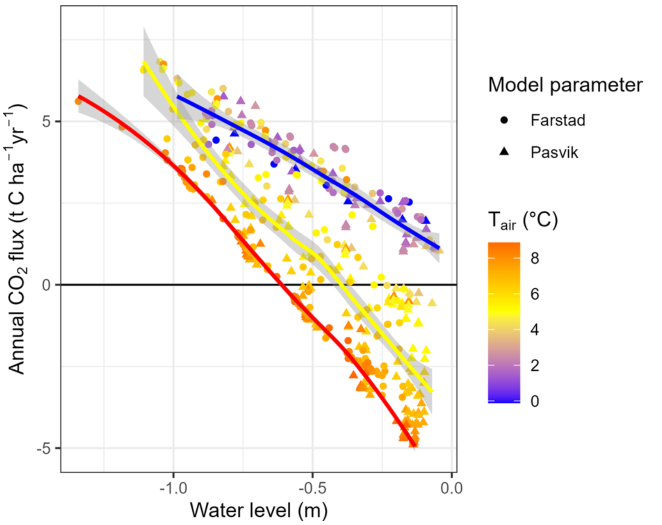

Fig. 6 | Simulated annual CO_2 budgets at the 50 sites across Norway under the climate conditions during 2001–2022. Water level gradients represent 5 different drainage depths (−1, −0.75, −0.5, −0.25, and 0 m). Colors indicate the mean annual temperature (T_{air}). To aid the visualization, smooth lines are added for three temperature groups (blue: <4 °C, yellow: ≥4 and <8 °C, red: ≥8 °C). The smooth lines are generated using a LOESS function and the gray bands are the 95% confidence intervals. Round points indicate simulations using the parameters calculated with Farstad field data and triangles indicate those that are based on Pasvik data.

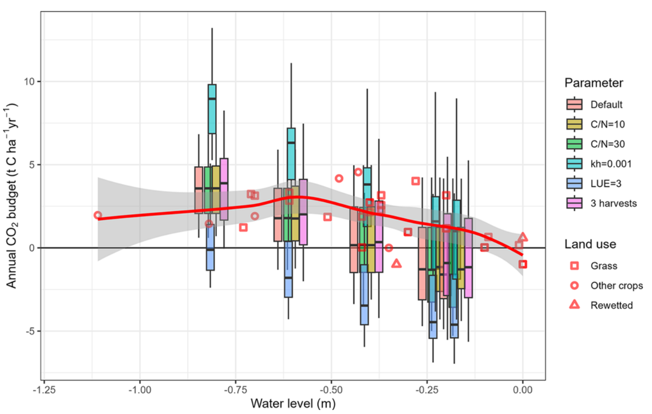

Fig. 7 | Simulated annual CO_2 flux budgets (NEE) at the 50 sites across Norway under the climate condition during 2001–2022. The colors of the boxes indicate different parameter settings (see Table 1). Red symbols are observed flux values obtained from other studies on cool temperate/boreal peatlands in Europe that were cultivated with grass (square), other crops (circle) and rewetted (triangle). A LOESS trend line (red) with 95% confidence interval bands is included to enhance the visualization of patterns in the data from other studies.

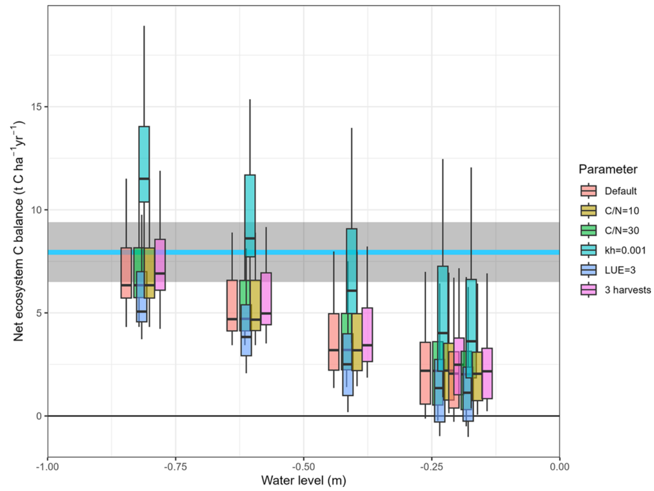

Fig. 8 | Simulated net ecosystem C balance (NECB) at the 50 sites across Norway under the climate conditions during 2001–2022. The colors of the boxes indicate different parameter settings (see Table 1). The blue horizontal lines indicate the Tier-1 EFs with the gray bands showing the 95% CI. The NECB includes net CO_2 exchange and harvested biomass export.

Table 2 | Summary of net ecosystem C balance (NECB, t C ha^{-1} yr^{-1}) from drained (water level: < −0.3 m) cropland on organic soils based on model simulations on the 50 sites, measurements from plots from Farstad in 2022–2024, and the dry and medium plots in Pasvik in 2022, compared to the Tier 1 EFs

Site – plot	Water level (m)	NECB		
		Mean	95% CI	n
50 sites	<−0.3 and > −0.5	4.2	[4.0, 4.4]	600[a]
	≤ −0.5 and > −0.7	6.0	[5.8, 6.2]	600
	≤ −0.7	7.7	[7.5, 7.9]	600
Pasvik - dry	−1.1	8.8	[6.1, 11.4]	3
Pasvik - medium	−0.4	4.9	[3.2, 6.5]	3
Tier 1 EF	<-0.3	7.9	[6.5, 9.4]	39

[a]Simulations with varying parameter values (see Table 1) are included.

variation in climate and site-specific hydrological conditions across Norway, we argue that this EF is unlikely to represent national average emissions accurately and may significantly overestimate CO_2 outputs, particularly in western coastal regions, where high precipitation frequently leads to temporary water saturation of drained sites.

Tier 1 EFs for drained organic soils are estimated based on observations from studies worldwide in the corresponding climate zone and land-use type[22]. These Tier 1 EFs can be associated with large uncertainties. For example, a study by He and Roulet[20] indicated that the EF overestimates CO_2 emissions from peatlands used for peat extraction in eastern Canada by ~50%. Summarizing all available field observations in Canada, Strack, et al.[23] concluded the mean CO_2 emission to be 35% lower than the IPCC Tier 1 EF. Similarly, Alm, et al.[24] suggested a CO_2 EF that is 28% lower than the Tier-1 EF for croplands on organic soils in Finland. It is noteworthy that boreal and temperate regions share the same Tier 1 CO_2 EFs for croplands on organic soils, even though this standard was established with more temperate sites than boreal ones[10]. By the IPCC classification[25], the Norwegian climate is classified as "cool temperate" in the majority of the land areas (Fig. S1). For countries in the same "cool temperate" region, the UK adopted a country-specific CO_2 EF (Tier 2) that is 9% smaller than the Tier 1 EF[26], while Germany adopted an EF that is 16% greater than the Tier 1 EF[27]. It is known that decomposition of SOC is generally greater in warmer regions[28]. Therefore, we suggest that the inaccuracies in Tier 1 estimations are largely caused by two factors: the coarse climate zone classification and regional biases in creating the EFs.

Moreover, drained peatlands are predominantly used for the cultivation of fodder crops in Europe. In Norway, for instance, approximately 85% of cultivated crops on peatlands are grasses[29], primarily timothy (*P. pratense*) and meadow fescue (*F. pratensis*). In fact, grasslands on drained organic soils have a distinct EF that is lower in CO_2 emissions (5.7 t CO_2-C ha^{-1} yr^{-1}) than those in croplands (see Table 2), although NEE is generally similar for the two types according to the observations from the literature (Fig. 5). By IPCC definitions[25], lands with forages (mainly grasses) are explicitly included in the cropland classification but can also be classified as grasslands. The studies that were used to derive the Tier 1 EFs for cropland also included sites with grass cultivation[10]. In Norway, (intensive) grasslands are defined as those that are regularly grazed but cannot be ploughed, while grasslands that can be ploughed are classified as cropland[9]. These ambiguities in land type definitions also contribute to the uncertainties in Tier 1 methodology. Nevertheless, it is known that different crop species come with diverse photosynthesis processes/capacities that determine the available soil C substrates and drive C decomposition[30]. Meanwhile, the crop type also determines harvest schemes and, thus, the amount of C being removed from an ecosystem as biomass, making it important in evaluating the sustainability of crop-specific peatland managements. Therefore, it is crucial to evaluate emissions associated with different crop groups, instead of broadly categorizing them into crops and grass. Furthermore, defining more land use types could also help reduce uncertainties. For example, abandoned peatlands are separated from the cropland and defined as an independent category with unique EFs in Denmark[31]. Other management practices, such as the plough frequency, which are important in different regions, also need to be considered for classification. Overall, Tier 1 EFs serve as a simple but rough way to estimate GHG emissions for countries without detailed land use and emissions data; however, to achieve finer classification and more accurate emission estimations, countries are urged to increase field monitoring to establish country-specific Tier 2 or 3 methodologies, wherever feasible. The Tier 2/3 methodologies establish a more suitable classification system for a specific country that can effectively avoid uncertainties in Tier 1 classifications for climate and land use types.

Following the IPCC definition, the Tier 1 CO_2 EF reflects the annual net change in SOC[10]. In this study, we used NECB as a proxy for SOC change to enable comparison with the Tier 1 EF[27]. NECB was calculated based on simulated NEE and harvested biomass, which are its main components. Our simulated NEE aligns well with observations from other cool temperate/boreal European sites (Fig. 7), and the modeled harvests are consistent with Norwegian agricultural statistics (Fig. S4), supporting the reliability of our NECB estimates. Although dissolved organic carbon (DOC) losses are minor in both our simulations (0.01 t C ha^{-1} yr^{-1}) and the IPCC EF (0.12 t C ha^{-1} yr^{-1}) and thus excluded from NECB estimates, they may be more relevant in Norway's high-precipitation regions and warrant further study. Additionally, we assumed no organic fertilization, as none was applied at our study sites; however, organic fertilizers are used on peatlands in some locations in Norway, potentially affecting C inputs and emissions, and should also be further investigated. Importantly, annual net SOC change is not an easily measurable variable, and is typically derived from field flux measurements or subsidence data, adjusted using a series of simple assumptions—such as constant litter inputs and autotrophic respiration[10]. Although direct estimates of SOC change are subject to considerable uncertainty, the strong agreement between our modeled NEE and harvest data with independent observations provides confidence in the reliability of our NECB-based approach for quantifying CO_2 emissions from cultivated peatlands.

Our results include simulations for higher water levels (> −0.3 m), reflecting the potential impact of raising water table. As expected, raising water tables significantly reduces the net CO_2 emission (or enhances the net uptake). We also found that the ecosystem CO_2 balance largely depends on the annual mean air temperature. This temperature effect is closely tied to the length of the growing season, suggesting that warmer regions (e.g., southern and western Norway) provide longer periods for photosynthetic CO_2 sequestration, which offsets the decomposition increase. The CoupModel also includes simulations of CH_4 and nitrous oxide (N_2O) fluxes (Fig. S5), suggesting raising water tables could instigate a notable rise in CH_4 emissions and a decline in N_2O emissions. Taking all three GHGs into account, a water level of ~ −0.45 m can minimize the overall emissions according to the model outputs. Further increasing the water level can increase GHG emissions as CH_4 emissions become more significant[14,32]. However, the simulated CH_4 emissions at water levels > −0.3 m are generally higher than emissions from rewetted peatlands (on average 3.7 (95% CI: 2.4, 4.9) t CO_2-equivalent C ha^{-1} yr^{-1})[33], suggesting our projected total emissions after raising water table might be overstated. On the other hand, a meta-analysis indicated that the effect of raising water table on CH_4 emissions has a temporal legacy, and the emissions could increase over years after raising water table[14]. A further increase in water level, which exceeds the ground surface, could also impose stress to plants and inhibit photosynthesis, even for species that are tolerant of inundation[34], and thus further dampen the GHG mitigation effects of raising water table together with the increased CH_4 production. The contribution of N_2O to total GHG is relatively small according to the simulations; however, to verify the result, more field data are needed and underlying processes are yet to be better understood[35,36]. Overall, while raising water tables is likely to reduce GHG emissions, considerable knowledge gaps still exist regarding the best practices to optimize its effects across various regions and sites.

Our model calibration was based on data collected from two sites in Norway. The sites represent typical land use (i.e., drainage implementation, soil C content and vegetation) and cover a wide range of conditions from dry/cold to wet/warm climate in Norway. After calibration, the model reproduced the flux dynamics at both sites with sufficient accuracy to be considered acceptable. Most under-/overestimations of the daily fluxes are primarily linked to discrepancies between the modeled and observed water table levels (Fig. S1). For example, in the medium water table plot at the Pasvik site, the model simulated a lower water table during the early growing season, which led to higher CO_2 emissions. Later in the season, the model predicted a higher water table than observed, resulting in an underestimation of CO_2 emissions. To evaluate the potential flux variations across the country, we conducted flux simulations using a diverse range of climatic conditions from 50 sites. While the mismatch of fluxes at the plot scale could introduce uncertainty in modeled water table levels across the 50 sites, it does not affect the underlying relationships between water table and fluxes/NECB, based on which we compared the NECB with the IPCC emission factors. These simulations also incorporated various values for key model parameters representing a variety of soil properties and harvest schemes. Among the parameters, the k_h, which indicates the decomposition rate for soil humus, showed a great influence on the outputs of CO_2 fluxes. By doubling the value of k_h from 0.0005 to 0.001 day^{-1}, CO_2 emissions increased by up to 3-fold due to a higher heterotrophic respiration[37]. The value of k_h depends on soil type, climate and land use; nevertheless, higher k_h values than 0.001 are unlikely, as they would result in simulated fluxes far greater than observed (Fig. 7). Thus, the emissions should not be greater than the flux outputs using this value. In contrast, changes in soil C/N ratio exhibit little effect on CO_2 emissions. While soil C content and k_h are expected to be correlated since higher C:N ratios generally correspond to slower decomposition, the limited sensitivity of flux outputs to C/N suggests that its influence is largely masked by k_h when parameters are tuned individually. As a result, adjusting k_h alone can compensate for variations in C/N, but could potentially introduce compensatory effects and bias in ensemble simulations. Future work should examine the empirical relationship between k_h and soil C/N in Norwegian peatlands and incorporate this dependency into parameterization to improve model realism. In addition, CO_2 fluxes are very sensitive to light use efficiency and the default value of 2 g MJ^{-1} is typical for timothy grasses[38], which are widely cultivated in Norway. Our results suggest that crops that have higher light use efficiency can significantly reduce the emissions of CO_2 from the field even in the well-drained sites. Harvest frequency does not show a significant effect on the CO_2 budget and this conclusion was also found in Nielsen, et al.[39]. This suggests that the temporary decline in photosynthetic CO_2 uptake following harvest has a negligible effect on the annual CO_2 budget. Overall, our CO_2 flux simulations, which were performed based on a variety of land use assumptions and climate conditions in Norway, generally align with observations from the literature. Nonetheless, applying parameter sets calibrated at only two sites to the national scale inevitably introduces substantial representational uncertainty. These uncertainties are further amplified by the limited availability of long-term, spatially distributed field measurements, which constrains our ability to fully validate model performance across Norway's diverse peatland regions.

While we explored a range of potential emissions by varying key parameters, additional sources of uncertainty remain, mostly due to limitations in the field data. For example, because we lacked data from snow-covered periods, the model calibration did not include the winter season. While Evans et al. (2021) suggested using a constant value of 0.015 t C ha^{-1} $month^{-1}$ for the non-growing season, our modeling results showed substantial variability, with an average flux of 0.15 ± 0.14 (SD) t C ha^{-1} month-across 50 sites from December to February. Simulating winter fluxes remains challenging because they are highly sensitive to small changes in environmental conditions (e.g., snow depth) and, more importantly, due to limited understanding of the underlying processes and a scarcity of field data. Future efforts to improve winter modeling will be crucial, particularly given the significant impacts of climate change on winter environmental conditions. In general, with field data of larger temporal and spatial coverages, future analyses that include probabilistic parameter sampling and error propagation from input datasets will result in a full quantitative uncertainty.

Conclusion

National GHG accounting is important for policymakers to understand the level and source of GHG emissions. The Tier 1 method is recommended by IPCC[22] for national GHG accounting in countries without sufficient field data, but the results are associated with large uncertainties. Our results from Norway suggest that countries that use Tier 1 EFs may potentially overestimate CO_2 emissions by 31–88% from cultivated peatlands in broad cool temperate/boreal regions with annual mean water table that is higher than –0.7 m. These inaccuracies in CO_2 estimates are mainly due to (1) coarse classifications for climate zones and land use types, (2) regional bias, and (3) a lack of inter-annual variations. The results can distort the mitigation potential of drained peatlands and reduce the effectiveness of mitigation measures.

In general, establishing a country-specific Tier 2/3 method is highly recommended, particularly for such significant C sources as drained peatlands. The Tier 2/3 method will greatly increase the accuracy in CO_2 estimates, considering large variabilities in crops and management regimes among different countries. It can help better understand the processes and sources resulting in emissions across the country, allowing for more accurate and targeted GHG mitigation efforts. The Tier 2 method adopts more accurate country specific EFs that are generated based on field GHG observations from typical cultivated peatland sites in a country[27]. A Tier 3 method also requires field data to calibrate processed-based models to produce more region-based GHG emission estimates. However, peatland field GHG observations are lacking in many countries around the globe including Norway[12]. Thus, more field observations, either on GHG fluxes or the key driver water table, should be prioritized as a crucial step for improving the peatland GHG accounting. This will also enable further refinement of the IPCC Tier 1 factors in the future, when more data is available.

Further, our field observations in Pasvik indicate that water table levels can exhibit significant spatial variation within a single drained field. Lower emissions were noted in wetter areas far away from the main drainage ditch. This highlights the importance of considering small scale or even site-specific conditions in reducing uncertainties of emission estimates and improving agricultural management. While CO_2 is the largest source (82% of the emissions based on today's Tier 1 methodology), both N_2O and CH_4 emissions can be important for agricultural land use and need to be investigated. Future model developments and inventory improvements must integrate all major gases to ensure a comprehensive evaluation of mitigation measures, especially as water-table manipulation can shift the relative contribution of different GHGs.

Overall, our national-scale simulations revealed several practical challenges, including information gaps in GHG flux, drainage, water table, and management data. Addressing these barriers is essential for improving the accuracy and policy relevance of national GHG accounting. Therefore, we strongly recommend establishing a well-distributed national monitoring network, strategically located across key peatland regions. A coordinated, long-term observational program would provide the empirical foundation needed to support more accurate emission factors, refine model parameterization, and ultimately strengthen Norway's capacity to implement Tier2/3 methodologies and design effective mitigation strategies. Future studies require collaboration among scientists, policy makers and stakeholders, combining more detailed management data with process-based models, to gain a more comprehensive understanding of GHG emissions from cultivated peatlands, which will allow for fine-tuning of potential mitigation measures.

Methods

Overview of field data from Farstad and Pasvik

For the model calibration, we used data collected at two cultivated peatland sites in different climatic regions of Norway. The first site is in Farstad, Hustadvika Municipality in western Norway (62° 57' 50"N, 7° 8' 17"E). Mean annual temperature and precipitation are 7.2 °C and 2230 mm, respectively. Temperatures throughout the year are relatively mild, ranging from a mean of 0.8 °C in January to 15.5 °C in July, with a growing season of 6–7 months. The site was originally drained in 1972 for forage production and in 1980, drainage pipes were installed at 8 m spacing and 1.3 m deep across the field, ending in a deep open ditch. The field has been sown with a timothy (*Phleum pratense*) based seed mixture. A detailed site description is available in Gunathilake, et al.[40]. Flux measurements of CO_2 were conducted in multiple campaigns between June 2022 and September 2024, with individual campaigns conducted between the months of February and October. Net ecosystem exchange (NEE) was measured on five replicate plots using manual transparent chambers and a LI-COR LI-850 CO_2 analyzer (LI-COR Environmental, Lincoln, NE, USA) in a custom-built casing and with a cooling system to minimize temperature variations during measurements. Ecosystem respiration (R_{eco}) was measured by placing opaque covers over the chambers. In total, 914 measurements of NEE and 870 measurements of R_{eco} were conducted. Gross primary production (GPP) was estimated as the difference between NEE and R_{eco}. The field was fertilized at the end of April/beginning of May and at the end of June/beginning of July for a total of 260 kg N ha^{-1} yr^{-1}. Grass was harvested twice per year, in late June and late August. Water table depth was measured hourly using a DST 22 pressure probe (SEBA Hydrometrie, Kaufbeuren, Germany). Hourly soil moisture and temperature at 7 cm and 30 cm depth were measured using Trime Pico 32 sensors (IMKO Micromodultechnik GmbH, Ettlingen, Germany). Due to high precipitation and low hydraulic conductivity, the mean annual water table is high at -0.24 m, but ranging from −0.82–0.06 m and generally high-water levels occur mostly outside the growing season (Fig. S1). Soil organic C content was 34 ± 1.4%, total N was 1.5 ± 0.13% (C/N ratio: 22), and bulk density was 0.23 ± 0.03 g cm^{-3} at 5-10 cm depth[40].

The second site used for model calibration is in the Pasvik Valley (69˚28'33"N, 29˚59'24"E) close to the NIBIO Svanhovd research station in northern Norway. The site has a growing season of 4–5 months. The annual mean temperature and precipitation are 0.5 °C and 480 mm, respectively. The mean temperatures range from −11 °C in January to 15 °C in July. The peatland was drained in the 1970s and cultivated with a mix of meadow fescue (*Festuca pratensis*) and timothy (*P. pratense*). The peat depth varies from 1 to 1.8 m, overlying sandy clay. Drainage pipes were laid across the field at 4–8 m intervals and a depth of 0.8 m, discharging into a deep open ditch. Depending on the distance to the drainage and water flow, water levels vary across different locations at the site. We established 5 plots (~ 50 m^2) along the water level gradient. In this study, we used data in 2022 from three plots that were ~100–150 m apart from each other. They had a high contrast in water table levels, with means of −1.10 ± 0.24 (SD), −0.47 ± 0.25, and −0.25 ± 0.17 m in 2022 that represent dry, medium, and wet conditions, respectively. The annual temperature and precipitation in 2022 were 1.7 and 506 mm, respectively, similar to the long-term averages. Soils at the site, on average, contain 46 ± 7 % C (organic), 2.8 ± 0.6% nitrogen (N) (C/N ratio: 16.5 ± 2.2) with a bulk density of 0.23 ± 0.03 g cm^{-3}. The soil pH is 6.2 ± 0.1. Fluxes of CO_2 were measured using automatic chambers[41] with a Picarro G2508 analyzer (Picarro, Inc., Santa Clara, CA, USA) at 8 h intervals throughout the growing season. Data from 3 chambers were used in each

Table 3 | Key parameters and their values used in model simulations for the 50 sites

Parameter	Range	Step	Default value	Farstad value	Pasvik value	Unit
C:N ratio	10–30	20	15	15	15	-
Light use efficiency	2–3	1	2	2	3	g MJ^{-1}
Rate coefficient for the decay of humus k_h	0.0005–0.001	0.0005	0.0005	0.001	0.0005	day^{-1}
Harvest	2–3	1	2	2	1	times yr^{-1}

plot. Due to power failure, data were missing between July 20 and August 10. Fertilization (Yara "18-3-15") was applied twice on May 12 and August 9 with a total of 126 kg N ha^{-1}. More information about the flux measurements is outlined in Zhao, et al.[42]. The grass was harvested once on August 9th and was cut at stubble height. Soil temperature and moisture were monitored at 2, 5, 10, 20, 50, and 75 cm depths in each plot using Stevens Water Hydraprobe (Stevens Water Monitoring Systems, Inc., Portland, OR, USA). Water levels in each plot were measured using Seametrics PT12 (Seametrics, Inc., Kent, WA, USA).

CoupModel description and calibration using data from Farstad and Pasvik

The CoupModel (https://www.coupmodel.com/) has a structure that allows switching among various simulation requirements. The model accounts for both heat and water conditions with complete process linkage between them and suitable boundary conditions to simulate snow and ice. It represents the soil profile through 20 compartments, ranging from a shallow 5 cm layer to a deep 1 m layer, covering soil up to 10 m deep. Water table dynamics are modeled based on pre-set ditch depths of the drainage, assuming water inputs are only from precipitation. Meteorological inputs are considered at a 2-meter height, with surface temperatures computed using an energy balance approach. A single plant layer is assumed in the model, with its physical size and coverage dynamically simulated based on plant biomass. This model links the plant's biomass above and below the soil surface with the physical conditions. The model uses hourly meteorological data, 7.5-min time steps for soil physical conditions, and 15-min time steps for plant conditions. C and N fluxes are modeled based on a light use efficiency approach simulating photosynthesis. Allocation to roots, leaves, and stems relies on phenological development and grain formation. The model considers two components in the decomposition of dead organic material, litter and humus, regulated by water and temperature conditions. For harvests, leaf biomass was partitioned as follows: 90% was removed, 5% remained as living tissue, and 5% was transferred to litter. For stem biomass, 70% was removed, 25% remained living, and 5% became litter. CO_2 production was modeled based on substrate availability. Default values for C/N ratios were used in the model initialization, set at 10 for humus and 25 for litter. Nitrogen inputs from fertilization, applied twice per growing season, were incorporated into the model simulations. Nitrogen dynamics were integrated throughout the simulation, influencing all processes related to organic matter decomposition and nutrient cycling.

The initial model setup was based on previous simulations from a spruce plantation on an agricultural peatland in the southwest of Sweden[43,44]. This basic setup was also used as a reference for simulation of other European treeless peatlands covering both agricultural land use and natural areas[45]. The settings for plant conditions were selected following those used by Senapati, et al.[46] on grasslands. The specific conditions at Farstad and Pasvik were used to calibrate the model in various steps. First, the hydrological conditions were simulated, and, at Pasvik, the simulations were compared with data of soil frost and snow depth (2002–2023) at a site of similar land use ~3 km away from the study site. Then, model simulations were performed by considering the hourly measured water levels, soil moisture, and temperature at the study sites. In this step, the snow and frost simulations were used to select the crucial parameters for primarily calibrating water level dynamics at Pasvik. The modeled water table dynamics at the two sites are present in Fig. S1 together with the measured values. The parameters (see Table S1 for the parameters and explanation) were calibrated via 15,000 Monte-Carlo based simulations to define an ensemble of accepted candidates. Various criteria and sizes of the ensembles were tested to minimize the discrepancy between simulated and measured CO_2 fluxes in the field. Model files with model inputs and parameter settings are archived on Zenodo (doi: 10.5281/zenodo.17978789, see Table S2 for the description of the files).

CoupModel simulations for 50 sites

CoupModel simulations were performed for 50 sites, which were selected to produce a good representation of overall climatic conditions for croplands on organic soils in Norway. The selection was separated into regions within which a random selection was chosen. The number of sites chosen in each of these regions was weighted by the total area of cultivated peat in each zone according to the organic soil and cultivated land distribution data from the AR5 classification system: http://hdl.handle.net/11250/2596511[47]. This approach ensured national coverage across diverse climate and solar conditions while keeping the 50 sites proportional to the overall distribution of cultivated peatlands. Hourly climate data during 2001–2022 were downloaded from Open-Meteo (open-meteo.com) at the locations of the 50 sites and used as drivers for model simulations. The variables include mean air temperature, global radiation, net radiation, relative humidity, precipitation, vapor pressure deficit, and wind speed. No field measurements were performed.

Two sets of simulations were run for the 50 sites, one each for the parameters tuned to the Farstad and Pasvik data. In addition, we also assessed potential variances of the model outputs in response to changes in soil properties and land management by using different values for key parameters (see Table 3) across a range of drainage depths (from −1 to 0 m). In the simulations, adjustments were made to one parameter at a time, while the remaining parameters were kept at their default values. The model version 6.3.1 from April 2025 was used for the simulations. Model files with model inputs and parameter settings are archived on Zenodo (doi: 10.5281/zenodo.13353432, see Table S2 for the description of the files). In addition, we also derived field observation data of CO_2 fluxes from synthesis studies on cultivated peatlands[13,48]. Data from 42 sites that were in latitudes higher than 55°N were used to represent European cool temperate and boreal regions (see Table S3 for the data) and compared with our modeled fluxes. According to IPCC[25], the majority of Norway is classified as cool temperate regions with boreal regions distributed in the northern inland areas (Fig. S2).

According to the IPCC definition, the Tier 1 CO_2 EF reflects changes in the soil organic C (SOC) pool[10]. To compare our simulations at the 50 sites with the EF, we calculated the net ecosystem C balance (NECB) as a sum of simulated NEE and C in harvested biomass. NECB at Pasvik was also estimated using the measured flux and biomass harvest data. Specifically, annual NEE was calculated using the mean measured flux during the growing season (4.5 months), while a CO_2 flux of 1.5 g C m^{-2} $month^{-1}$ was assumed for the non-growing season following Evans, et al.[13]. NECB was not derived from field observations at Farstad because the sparse CO_2 flux measurements could lead to highly biased estimates of the C budget. To align with the IPCC definition of drained conditions[10], we compared the NECB values under an annual mean water table level below –0.3 m with the Tier 1 EF for CO_2. In this study, we use negative values to indicate ecosystem C uptake and positive values for emissions.

Data availability

Model files with observation data, model inputs and parameter settings are archived on Zenodo (https://doi.org/10.5281/zenodo.13353432, see Table S2 for the description of the files).

Code availability

All code used to generate the figures is archived on Zenodo (https://doi.org/10.5281/zenodo.19018379).

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Author contributions
J.Z., M.T., P.J., E.J., G.S., and J.Y. contributed to the conceptualization and methodology of the study. J.Z., M.T., M.M., E.F., C.K., D.K. and R.K. contributed to the installation, maintenance and coordination of field measurements. J.Z., M.T., E.J., M.M., S.W., K.F., K.B., J.R., and C.M. contributed to the data curation, analysis, and visualization. J.Z., M.T., and P.J. made model calibrations/simulations. J.Z. led the manuscript writing, and all the authors contributed critically to the final manuscript.

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Competing interests
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